

Course 5302

5 Credits: 4

Program Core

Source-grounded

Data Mining

To understand data warehouse concepts, architecture, business analysis and tools

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 5302 is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5302.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Robotic Process Automation
Course code	5302
Course title	Data Mining
Semester	5 Credits: 4
Course category	Program Core
Periods per week	4 (L: 4 T: 0 P: 0) Periods per semester: 60

Item	Official information
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

32 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To understand data warehouse concepts, architecture, business analysis and tools
- To understand data pre-processing and data visualization techniques
- To study algorithms for finding hidden and interesting patterns in data
- To understand and apply various classification and clustering techniques using tools.

Course outcomes

CO	Official outcome	Study level
CO1	12 Understanding Warehousing Apply appropriate techniques to convert raw data into	See official syllabus
CO2	suitable format for practical data mining tasks and 18 Applying understand different Classification Models. Analyze and compare various classification algorithms	See official syllabus
CO3	15 Applying and apply in appropriate domain. Make use of the concept of association rule mining in real	See official syllabus
CO4	world scenario. Select appropriate clustering and 13 Applying algorithms for various applications Course Outcomes:	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3 2 1 2
CO2	CO2 3 2 2 2
CO3	CO3 3 2 2 2
CO4	CO4 3 2 2 2

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Understand the core concepts of Data Mining and Data Warehousing	9	As specified
Module 2	:	10	As specified
Module 3	:	6	As specified
Module 4	:	7	As specified

MODULE 1

Understand the core concepts of Data Mining and Data Warehousing

This module is presented from the official syllabus scope for Data Mining. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Understand the core concepts of Data Mining and Data Warehousing
- Official duration: As specified
- Official topic entries captured: 9
- The full source excerpt is preserved below for coverage checking.

Data Mining Concepts and Applications

Data Mining Concepts and Applications is an official topic in Module 1 of Data Mining. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Data Mining Concepts and Applications using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, data mining concepts and applications should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Data Mining Concepts and Applications means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-Data Mining Concepts and Applications

മലയാളത്തിൽ നിർവ്വചന/അർത്ഥം, ഘട്ടങ്ങൾ, പ്രയോഗം എന്നിവയെക്കുറിച്ച് വിശദീകരിക്കുക. Technical terms English-ൽ നിർവ്വചിക്കുക.

Data Mining Stages

Data Mining Stages is an official topic in Module 1 of Data Mining. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Data Mining Stages using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, data mining stages should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Data Mining Stages means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-Data Mining Stages ഡാറ്റാ മൈനിംഗ് സ്റ്റേജ്
definition/meaning ഡാറ്റാ മൈനിംഗ് ഡാറ്റാ മൈനിംഗ്, steps, application
ഡാറ്റാ മൈനിംഗ് ഡാറ്റാ മൈനിംഗ്. Technical terms English-Data Mining Stages

Data Mining Models

Data Mining Models is an official topic in Module 1 of Data Mining. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Data Mining Models using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, data mining models should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Data Mining Models means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Data Mining Models ഓരോന്നിനും നിങ്ങളുടെ നിർവ്വചനം/അർത്ഥം നിർവ്വചിക്കുക. ഓരോന്നിനും ഓരോന്നിനും, steps, application ഓരോന്നിനും നിങ്ങളുടെ നിർവ്വചനം നിർവ്വചിക്കുക. Technical terms English-ൽ നിങ്ങളുടെ നിർവ്വചനം നിർവ്വചിക്കുക.

Applications of Data Mining

Applications of Data Mining is an official topic in Module 1 of Data Mining. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Applications of Data Mining using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, applications of data mining should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Applications of Data Mining means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-എന്ന ആപ്ലിക്കേഷൻ ഓഫ് ഡാറ്റാ മൈനിംഗ് ഏതെങ്കിലും
ഒരു നിർവ്വചനം/അർത്ഥം ഉൾക്കൊള്ളുന്നു. ഏതെങ്കിലും ഏതെങ്കിലും ഏതെങ്കിലും, steps,
application ഏതെങ്കിലും ഏതെങ്കിലും ഏതെങ്കിലും. Technical terms English-ൽ എഴുതുക
ഏതെങ്കിലും.

Data warehouse and its architecture

Data warehouse and its architecture is an official topic in Module 1 of Data Mining. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Data warehouse and its architecture using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, data warehouse and its architecture should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Data warehouse and its architecture means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Data warehouse and its architecture

Data warehouse definition/meaning. Data warehouse architecture, steps, application. Technical terms English-Malayalam.

OLAP vs OLTP

OLAP vs OLTP is an official topic in Module 1 of Data Mining. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify OLAP vs OLTP using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, olap vs oltp should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What OLAP vs OLTP means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-OLAP vs OLTP എന്നിങ്ങനെയുള്ള പദങ്ങൾ
definition/meaning എങ്ങനെ എഴുതണം. പദങ്ങൾ എങ്ങനെ എഴുതണം, steps, application
എങ്ങനെ എഴുതണം എങ്ങനെ. Technical terms English-എങ്ങനെ എഴുതണം.

OLAP operations 2 Understanding Schemas of multidimensional Data model-Star

OLAP operations 2 Understanding Schemas of multidimensional Data model-Star is an official topic in Module 1 of Data Mining. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify OLAP operations 2 Understanding Schemas of multidimensional Data model-Star using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, olap operations 2 understanding schemas of multidimensional data model-star should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What OLAP operations 2 Understanding Schemas of multidimensional Data model-Star means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- OLAP operations 2 Understanding Schemas of multidimensional Data model-Star schema definition/meaning. **English:** definition/meaning, steps, application. Technical terms English- Malayalam.

2 Understanding schema, Snowflake schema and Fast Constellation,

2 Understanding schema, Snowflake schema and Fast Constellation, is an official topic in Module 1 of Data Mining. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding schema, Snowflake schema and Fast Constellation, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding schema, snowflake schema and fast constellation, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding schema, Snowflake schema and Fast Constellation, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 1-2 Understanding schema, Snowflake schema and Fast Constellation, തിരഞ്ഞെടുക്കൽ നിർവ്വചന/അർത്ഥം തിരഞ്ഞെടുക്കൽ നിർവ്വചനം. തിരഞ്ഞെടുക്കൽ നിർവ്വചനം, steps, application തിരഞ്ഞെടുക്കൽ നിർവ്വചനം തിരഞ്ഞെടുക്കൽ നിർവ്വചനം. Technical terms English-ൽ തിരഞ്ഞെടുക്കൽ നിർവ്വചനം തിരഞ്ഞെടുക്കൽ നിർവ്വചനം.

Applications of DWH 1 Understanding Introduction to Data Mining - Data Mining Concepts and Applications, Data Mining Stages, Data Mining Models, Applications of Data Mining, Data Warehousing (DWH) and On-Line Analytical Processing (OLAP), OLAP vs OLTP, OLAP operations, Schemas of multidimensional Data model-Star schema, Snowflake schema and Fast Constellation, Applications of DWH . Understanding and Apply various Data Preprocessing steps and

Applications of DWH 1 Understanding Introduction to Data Mining – Data Mining Concepts and Applications, Data Mining Stages, Data Mining Models, Applications of Data Mining, Data Warehousing (DWH) and On-Line Analytical Processing (OLAP), OLAP vs OLTP, OLAP operations, Schemas of multidimensional Data model-Star schema, Snowflake schema and Fast Constellation, Applications of DWH . Understanding and Apply various Data Preprocessing steps and is an official topic in Module 1 of Data Mining. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Applications of DWH 1 Understanding Introduction to Data Mining – Data Mining Concepts and Applications, Data Mining Stages, Data Mining Models, Applications of Data Mining, Data Warehousing (DWH) and On-Line Analytical Processing (OLAP), OLAP vs OLTP, OLAP operations, Schemas of multidimensional Data model-Star schema, Snowflake schema and Fast Constellation, Applications of DWH . Understanding and Apply various Data Preprocessing steps and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, applications of dwh 1 understanding introduction to data mining – data mining concepts and applications, data mining stages, data mining models,

applications of data mining, data warehousing (dwh) and on-line analytical processing (olap), olap vs oltp, olap operations, schemas of multidimensional data model-star schema, snowflake schema and fast constellation, applications of dwh . understanding and apply various data preprocessing steps and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Applications of DWH 1 Understanding Introduction to Data Mining – Data Mining Concepts and Applications, Data Mining Stages, Data Mining Models, Applications of Data Mining, Data Warehousing (DWH) and On-Line Analytical Processing (OLAP), OLAP vs OLTP, OLAP operations, Schemas of multidimensional Data model-Star schema, Snowflake schema and Fast Constellation, Applications of DWH . Understanding and Apply various Data Preprocessing steps and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-എന്ന ആപ്ലിക്കേഷൻ ഓഫ് ഡി.എച്ച്.ടി. 1. Understanding Introduction to Data Mining – Data Mining Concepts and Applications, Data Mining Stages, Data Mining Models, Applications of Data Mining, Data Warehousing (DWH) and On-Line Analytical Processing (OLAP), OLAP vs OLTP, OLAP operations, Schemas of multidimensional Data model-Star schema, Snowflake schema and Fast Constellation, Applications of DWH . Understanding and Apply various Data Preprocessing steps and ഏകദേശം definition/meaning എന്തെങ്കിലും. ഏകദേശം ഏകദേശം ഏകദേശം, steps, application എന്തെങ്കിലും ഏകദേശം. Technical terms English-എന്ന ആപ്ലിക്കേഷൻ ഓഫ് ഡി.എച്ച്.ടി.എന്ന ആപ്ലിക്കേഷൻ ഓഫ് ഡി.എച്ച്.ടി.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

Understand the core concepts of Data Mining and Data Warehousing

M1.01	Data Mining Concepts and Applications	1
Understanding		
M1.02	Data Mining Stages	1
Understanding		
M1.03	Data Mining Models	1
Understanding		
M1.04	Applications of Data Mining	1
Understanding		
M1.05	Data warehouse and its architecture	2
Understanding		
M1.06	OLAP vs OLTP	1
Understanding		
M1.07	OLAP operations	2
Understanding		
M1.08	Schemas of multidimensional Data model-Star schema, Snowflake schema and Fast Constellation,	2
Understanding		

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

⋮

This module is presented from the official syllabus scope for Data Mining. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: :
- Official duration: As specified
- Official topic entries captured: 10
- The full source excerpt is preserved below for coverage checking.

Data Preprocessing Concepts 1 Understanding.

Data Preprocessing Concepts 1 Understanding. is an official topic in Module 2 of Data Mining. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Data Preprocessing Concepts 1 Understanding. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, data preprocessing concepts 1 understanding. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Data Preprocessing Concepts 1 Understanding. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- Data Preprocessing Concepts 1 Understanding.
Data Preprocessing definition/meaning. Data Preprocessing steps, application. Technical terms English- Malayalam.
Data Preprocessing.

Data Cleaning

Data Cleaning is an official topic in Module 2 of Data Mining. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Data Cleaning using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, data cleaning should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Data Cleaning means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-Data Cleaning ഡാറ്റാ ക്ലീനിംഗ് ഡാറ്റാ
definition/meaning ഡാറ്റാ ക്ലീനിംഗ്. ഡാറ്റാ ക്ലീനിംഗ് ഡാറ്റാ, steps, application
ഡാറ്റാ ക്ലീനിംഗ് ഡാറ്റാ. Technical terms English-Data Cleaning

Data Integration 1 Applying Data Reduction Strategies Overview-Dimensionality

Data Integration 1 Applying Data Reduction Strategies Overview-Dimensionality is an official topic in Module 2 of Data Mining. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Data Integration 1 Applying Data Reduction Strategies Overview-Dimensionality using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, data integration 1 applying data reduction strategies overview-dimensionality should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Data Integration 1 Applying Data Reduction Strategies Overview-Dimensionality means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-ഡാറ്റാ ഇന്റഗ്രേഷൻ 1 Applying Data Reduction Strategies Overview-Dimensionality റിഡക്ഷൻ definition/meaning റിഡക്ഷൻ റിഡക്ഷൻ. റിഡക്ഷൻ റിഡക്ഷൻ, steps, application റിഡക്ഷൻ റിഡക്ഷൻ. Technical terms English-ഡാറ്റാ ഇന്റഗ്രേഷൻ.

2 Understanding. Reduction, Numerosity Reduction Data Transformation Strategies Overview-smoothing,

2 Understanding. Reduction, Numerosity Reduction Data Transformation Strategies Overview-smoothing, is an official topic in Module 2 of Data Mining. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding. Reduction, Numerosity Reduction Data Transformation Strategies Overview-smoothing, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding. reduction, numerosity reduction data transformation strategies overview-smoothing, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding. Reduction, Numerosity Reduction Data Transformation Strategies Overview-smoothing, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-2 Understanding. Reduction, Numerosity Reduction Data Transformation Strategies Overview-smoothing, ഏകീകരണം definition/meaning ഏകീകരണം. ഏകീകരണം ഏകീകരണം, steps, application ഏകീകരണം ഏകീകരണം. Technical terms English-ഏകീകരണം ഏകീകരണം.

2 Understanding. Attribute construction, Aggregation Data Transformation Strategies Overview -

2 Understanding. Attribute construction, Aggregation Data Transformation Strategies Overview - is an official topic in Module 2 of Data Mining. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding. Attribute construction, Aggregation Data Transformation Strategies Overview - using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding. attribute construction, aggregation data transformation strategies overview - should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding. Attribute construction, Aggregation Data Transformation Strategies Overview - means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-പാഠ്യ 2 Understanding. Attribute construction, Aggregation Data Transformation Strategies Overview - പരമ്പരപ്പെടുത്തലിന്റെ പരിധി definition/meaning പരമ്പരപ്പെടുത്തലിന്റെ പരിധി. പരമ്പരപ്പെടുത്തലിന്റെ പരിധി, steps, application പരമ്പരപ്പെടുത്തലിന്റെ പരിധി. Technical terms English-പാഠ്യ പരമ്പരപ്പെടുത്തലിന്റെ പരിധി.

Normalization- min-max normalization, z-score 2 Applying normalization Data Transformation Strategies Overview -

Normalization- min-max normalization, z-score 2 Applying normalization Data Transformation Strategies Overview - is an official topic in Module 2 of Data Mining. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Normalization- min-max normalization, z-score 2 Applying normalization Data Transformation Strategies Overview - using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, normalization- min-max normalization, z-score 2 applying normalization data transformation strategies overview - should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Normalization- min-max normalization, z-score 2 Applying normalization Data Transformation Strategies Overview - means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-നോർമലൈസേഷൻ Normalization- min-max normalization, z-score
2 Applying normalization Data Transformation Strategies Overview - നോർമലൈസേഷൻ
നോർമലൈസേഷൻ definition/meaning നോർമലൈസേഷൻ. നോർമലൈസേഷൻ നോർമലൈസേഷൻ, steps,
application നോർമലൈസേഷൻ നോർമലൈസേഷൻ നോർമലൈസേഷൻ. Technical terms English-നോർമലൈസേഷൻ
നോർമലൈസേഷൻ.

Discretization, Concept Hierarchy Generation 2 Understanding Introduction to Classification and Prediction, Issues

Discretization, Concept Hierarchy Generation 2 Understanding Introduction to Classification and Prediction, Issues is an official topic in Module 2 of Data Mining. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Discretization, Concept Hierarchy Generation 2 Understanding Introduction to Classification and Prediction, Issues using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, discretization, concept hierarchy generation 2 understanding introduction to classification and prediction, issues should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Discretization, Concept Hierarchy Generation 2 Understanding Introduction to Classification and Prediction, Issues means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- $\square\square$ Discretization, Concept Hierarchy Generation 2 Understanding Introduction to Classification and Prediction, Issues $\square\square\square\square\square\square\square\square\square$ $\square\square\square\square$ definition/meaning $\square\square\square\square\square\square\square\square\square$. $\square\square\square\square$ $\square\square\square\square$ $\square\square\square\square$, steps, application $\square\square\square\square$ $\square\square\square\square\square\square\square$ $\square\square\square\square$. Technical terms English- $\square$ $\square\square\square\square$ $\square\square\square\square\square\square\square\square$.

1 Understanding regarding classification and prediction

1 Understanding regarding classification and prediction is an official topic in Module 2 of Data Mining. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding regarding classification and prediction using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding regarding classification and prediction should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding regarding classification and prediction means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- 1 Understanding regarding classification and prediction **പ്രവർത്തനം** definition/meaning **പ്രവർത്തനം**. **പ്രവർത്തനം** **പ്രവർത്തനം**, steps, application **പ്രവർത്തനം** **പ്രവർത്തനം**. **Technical terms** English- **പ്രവർത്തനം** **പ്രവർത്തനം**.

Decision Tree- ID3

Decision Tree- ID3 is an official topic in Module 2 of Data Mining. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Decision Tree- ID3 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, decision tree- id3 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Decision Tree- ID3 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-Decision Tree- ID3 algorithm definition/meaning. algorithm steps, application. Technical terms English-Decision Tree.

Naive Bayes Classifier. 3 Applying Data Preprocessing: Data Preprocessing Concepts, Data Cleaning, Data integration and transformation, Data Reduction Strategies-Dimensionality Reduction, Numerosity Reduction, Data Transformation Strategies Overview-smoothing, Attribute construction, Aggregation, Normalization- min-max normalization, z-score normalization, Discretization, Concept Hierarchy Generation Classification Models: Introduction to Classification and Prediction, Issues regarding classification and prediction, Decision Tree- ID3, Naive Bayes Classifier. Analyze and compare various classification algorithms and apply in

Naive Bayes Classifier. 3 Applying Data Preprocessing: Data Preprocessing Concepts, Data Cleaning, Data integration and transformation, Data Reduction Strategies-Dimensionality Reduction, Numerosity Reduction, Data Transformation Strategies Overview-smoothing, Attribute construction, Aggregation, Normalization- min-max normalization, z-score normalization, Discretization, Concept Hierarchy Generation Classification Models: Introduction to Classification and Prediction, Issues regarding classification and prediction, Decision Tree- ID3, Naive Bayes Classifier. Analyze and compare various classification algorithms and apply in is an official topic in Module 2 of Data Mining. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Naive Bayes Classifier. 3 Applying Data Preprocessing: Data Preprocessing Concepts, Data Cleaning, Data integration and transformation, Data Reduction Strategies-Dimensionality Reduction, Numerosity Reduction, Data Transformation Strategies Overview-smoothing, Attribute construction, Aggregation, Normalization- min-max normalization, z-score normalization, Discretization, Concept Hierarchy Generation Classification Models: Introduction to Classification and Prediction, Issues regarding classification and prediction, Decision Tree- ID3, Naive Bayes Classifier. Analyze and compare various classification algorithms and apply in using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.

- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, naive bayes classifier. 3 applying data preprocessing: data preprocessing concepts, data cleaning, data integration and transformation, data reduction strategies-dimensionality reduction, numerosity reduction, data transformation strategies overview-smoothing, attribute construction, aggregation, normalization- min-max normalization, z-score normalization, discretization, concept hierarchy generation classification models: introduction to classification and prediction, issues regarding classification and prediction, decision tree- id3, naive bayes classifier. analyze and compare various classification algorithms and apply in should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Naive Bayes Classifier. 3 Applying Data Preprocessing: Data Preprocessing Concepts, Data Cleaning, Data integration and transformation, Data Reduction Strategies-Dimensionality Reduction, Numerosity Reduction, Data Transformation Strategies Overview-smoothing, Attribute construction, Aggregation, Normalization- min-max normalization, z-score normalization, Discretization, Concept Hierarchy Generation Classification Models: Introduction to Classification and Prediction, Issues regarding classification and prediction, Decision Tree- ID3, Naive Bayes Classifier. Analyze and compare various classification algorithms and apply in means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Naive Bayes Classifier. 3 Applying Data Preprocessing: Data Preprocessing Concepts, Data Cleaning, Data integration and transformation, Data Reduction Strategies-Dimensionality Reduction, Numerosity Reduction, Data Transformation Strategies Overview-smoothing, Attribute construction, Aggregation, Normalization- min-max normalization, z-score normalization, Discretization, Concept Hierarchy Generation Classification Models: Introduction to Classification and Prediction, Issues regarding classification and prediction, Decision Tree- ID3, Naive Bayes Classifier. Analyze and compare various

classification algorithms and apply in രണ്ടാം വർഷം definition/meaning രണ്ടാം വർഷം. രണ്ടാം വർഷം രണ്ടാം വർഷം, steps, application രണ്ടാം വർഷം രണ്ടാം വർഷം. Technical terms English-രണ്ടാം വർഷം രണ്ടാം വർഷം.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

:			
Classification Models			
M2.01	Data Preprocessing Concepts Understanding.	1	
M2.02	Data Cleaning	1	Applying
M2.03	Data Integration	1	Applying
M2.04	Data Reduction Strategies Overview-Dimensionality Understanding.	2	
M2.05	Reduction, Numerosity Reduction Data Transformation Strategies Overview-smoothing, Understanding.	2	
M2.06	Attribute construction, Aggregation Data Transformation Strategies Overview - Normalization- min-max normalization, z-score normalization	2	Applying
M2.07	Data Transformation Strategies Overview - Discretization, Concept Hierarchy Generation Understanding	2	
M2.08	Introduction to Classification and Prediction, Issues Understanding	1	

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

:

This module is presented from the official syllabus scope for Data Mining. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: :
- Official duration: As specified
- Official topic entries captured: 6
- The full source excerpt is preserved below for coverage checking.

2 Understanding characteristics

2 Understanding characteristics is an official topic in Module 3 of Data Mining. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding characteristics using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding characteristics should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding characteristics means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- 2 Understanding characteristics $\square\square\square\square\square\square\square\square$
 $\square\square\square\square$ definition/meaning $\square\square\square\square\square\square\square\square$. $\square\square\square\square$ $\square\square\square\square$ $\square\square\square\square$, steps,
application $\square\square\square\square$ $\square\square\square\square\square\square$ $\square\square\square\square$. Technical terms English- $\square$ $\square\square\square\square$
 $\square\square\square\square\square\square\square$.

Back propagation Algorithm.

Back propagation Algorithm. is an official topic in Module 3 of Data Mining. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Back propagation Algorithm. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, back propagation algorithm. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Back propagation Algorithm. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- $\square$ Back propagation Algorithm. $\square\square\square\square\square\square\square\square\square\square$
 $\square\square\square\square$ definition/meaning $\square\square\square\square\square\square\square\square\square\square$. $\square\square\square\square\square$ $\square\square\square\square\square$ $\square\square\square\square\square\square$, steps,
 application $\square\square\square\square$ $\square\square\square\square\square\square\square\square$ $\square\square\square\square\square$. Technical terms English- $\square$ $\square\square\square\square$
 $\square\square\square\square\square\square\square\square\square$.

Support Vector Machines 2 Understanding.

Support Vector Machines 2 Understanding. is an official topic in Module 3 of Data Mining. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Support Vector Machines 2 Understanding. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, support vector machines 2 understanding. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Support Vector Machines 2 Understanding. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Support Vector Machines 2 Understanding.

Support Vector Machines definition/meaning. Support Vector Machines, steps, application. Technical terms English- Malayalam.

Lazy Learners-K Nearest Neighbor Classifier.

Lazy Learners-K Nearest Neighbor Classifier. is an official topic in Module 3 of Data Mining. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Lazy Learners-K Nearest Neighbor Classifier. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, lazy learners-k nearest neighbor classifier. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Lazy Learners-K Nearest Neighbor Classifier. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- ലാസ്റ്റ് ലേണേഴ്സ്-ക നേരേസ്റ്റ് നെighbor ക്ലാസ്സൈറ്റ.

ലാസ്റ്റ് ലേണേഴ്സ് ക്ലാസ്സൈറ്റ definition/meaning ലാസ്റ്റ് ലേണേഴ്സ് ക്ലാസ്സൈറ്റ. ലാസ്റ്റ് ലേണേഴ്സ് ക്ലാസ്സൈറ്റ, steps, application ലാസ്റ്റ് ലേണേഴ്സ് ക്ലാസ്സൈറ്റ. Technical terms English-ലാസ്റ്റ് ലേണേഴ്സ് ക്ലാസ്സൈറ്റ.

Accuracy and error Measuresevaluation.

Accuracy and error Measuresevaluation. is an official topic in Module 3 of Data Mining. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Accuracy and error Measuresevaluation. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, accuracy and error measuresevaluation. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Accuracy and error Measuresevaluation. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Accuracy and error Measuresevaluation.

Accuracy definition/meaning . Accuracy steps, application . Technical terms English- Malayalam .

Prediction:-Linear Regression 3 Applying Artificial Neural Network-concepts and Characteristics, Back propagation Algorithm. Support Vector Machines, Lazy Learners-K Nearest Neighbor Classifier. Accuracy and error Measures evaluation. Prediction:-Linear Regression Make use of the concept of association rule mining in real world scenario.

Prediction:-Linear Regression 3 Applying Artificial Neural Network-concepts and Characteristics, Back propagation Algorithm. Support Vector Machines, Lazy Learners-K Nearest Neighbor Classifier. Accuracy and error Measures evaluation. Prediction:-Linear Regression Make use of the concept of association rule mining in real world scenario. is an official topic in Module 3 of Data Mining. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Prediction:-Linear Regression 3 Applying Artificial Neural Network-concepts and Characteristics, Back propagation Algorithm. Support Vector Machines, Lazy Learners-K Nearest Neighbor Classifier. Accuracy and error Measures evaluation. Prediction:-Linear Regression Make use of the concept of association rule mining in real world scenario. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, prediction:-linear regression 3 applying artificial neural network-concepts and characteristics, back propagation algorithm. support vector machines, lazy learners-k nearest neighbor classifier. accuracy and error measures evaluation. prediction:-linear regression make use of the concept of association rule mining in real world scenario. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Prediction:-Linear Regression 3 Applying Artificial Neural Network-concepts and Characteristics, Back propagation Algorithm. Support Vector Machines, Lazy Learners-K Nearest Neighbor Classifier. Accuracy and error Measures evaluation. Prediction:-Linear Regression Make use of the concept of association rule mining in real world scenario. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-പ്രവചനം:-ലിനിയർ റിഗ്രെഷൻ 3 പ്രയോജനപ്പെടുത്തുന്ന കൃത്രിമ ന്യൂറൽ നെറ്റ്-കോൺസെപ്റ്റുകൾ, ബാക്ക് പ്രൊപ്പഗേഷൻ അൾഗരിതം. സപ്പോർട്ട് വെക്ടർ മെഷീനുകൾ, ലാസി ലേണർ-ക നേർത്ത നെർബർ ക്ലാസ്സിഫയർ. കൃത്യതയും തെറ്റിടകളുടെ മൂല്യനിർണ്ണയവും. പ്രവചനം:-ലിനിയർ റിഗ്രെഷൻ പ്രയോജനപ്പെടുത്തുന്ന കോൺസെപ്റ്റ് ഓഫ് അസോസിയേഷൻ റൂൾ മൈനിംഗ് ഇൻ റീൽ വേൾഡ് സീനാറിയോ. പ്രവചനം:-ലിനിയർ റിഗ്രെഷൻ പ്രയോജനപ്പെടുത്തുന്ന കോൺസെപ്റ്റ് ഓഫ് അസോസിയേഷൻ റൂൾ മൈനിംഗ് ഇൻ റീൽ വേൾഡ് സീനാറിയോ. definition/meaning പ്രവചനം/അർത്ഥം. പ്രവചനം പ്രയോജനപ്പെടുത്തുന്ന കോൺസെപ്റ്റ് ഓഫ് അസോസിയേഷൻ റൂൾ മൈനിംഗ് ഇൻ റീൽ വേൾഡ് സീനാറിയോ, steps, application പ്രവചനം പ്രയോജനപ്പെടുത്തുന്ന കോൺസെപ്റ്റ് ഓഫ് അസോസിയേഷൻ റൂൾ മൈനിംഗ് ഇൻ റീൽ വേൾഡ് സീനാറിയോ. Technical terms English-പ്രവചനം പ്രയോജനപ്പെടുത്തുന്ന കോൺസെപ്റ്റ് ഓഫ് അസോസിയേഷൻ റൂൾ മൈനിംഗ് ഇൻ റീൽ വേൾഡ് സീനാറിയോ.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

:	appropriate domain.	
M3.01	Artificial Neural Networks- Concepts and Understanding characteristics	2
M3.02	Back propagation Algorithm. Applying	3
M3.03	Support Vector Machines Understanding.	2
M3.04	Lazy Learners-K Nearest Neighbor Classifier. Applying	3
M3.05	Accuracy and error Measuresevaluation. Applying	2
M3.06	Prediction:-Linear Regression Applying	3
Contents: Artificial Neural Network-concepts and Characteristics, Back propagation Algorithm. Support Vector Machines, Lazy Learners-K Nearest Neighbor Classifier. Accuracy		

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

:

This module is presented from the official syllabus scope for Data Mining. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: :
- Official duration: As specified
- Official topic entries captured: 7
- The full source excerpt is preserved below for coverage checking.

1 Apply confidence

1 Apply confidence is an official topic in Module 4 of Data Mining. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Apply confidence using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 apply confidence should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Apply confidence means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- 1 Apply confidence ഉറപ്പോടെ നിർവ്വചനം/അർത്ഥം ഉപയോഗിക്കുക. ഉപയോഗം, ഘട്ടങ്ങൾ, പ്രയോഗം ഉപയോഗിക്കുക. Technical terms English- ഉപയോഗിക്കുക.

FP-Growth Algorithm 3 Apply

FP-Growth Algorithm 3 Apply is an official topic in Module 4 of Data Mining. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify FP-Growth Algorithm 3 Apply using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, fp-growth algorithm 3 apply should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What FP-Growth Algorithm 3 Apply means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- $\square\square$ FP-Growth Algorithm 3 Apply $\square\square\square\square\square\square\square\square\square\square$
 $\square\square\square\square$ definition/meaning $\square\square\square\square\square\square\square\square\square\square$. $\square\square\square\square\square$ $\square\square\square\square\square$ $\square\square\square\square\square\square$, steps,
application $\square\square\square\square\square$ $\square\square\square\square\square\square\square\square$ $\square\square\square\square\square$. Technical terms English- $\square$ $\square\square\square\square$
 $\square\square\square\square\square\square\square\square$.

Cluster Analysis-Introduction, Concepts

Cluster Analysis-Introduction, Concepts is an official topic in Module 4 of Data Mining. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Cluster Analysis-Introduction, Concepts using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, cluster analysis-introduction, concepts should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Cluster Analysis-Introduction, Concepts means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-കുറ്റ Cluster Analysis-Introduction, Concepts
കുറ്റകുറ്റകുറ്റകുറ്റകുറ്റകുറ്റ definition/meaning കുറ്റകുറ്റകുറ്റകുറ്റകുറ്റകുറ്റ. കുറ്റകുറ്റ കുറ്റകുറ്റ കുറ്റകുറ്റകുറ്റ,
steps, application കുറ്റകുറ്റ കുറ്റകുറ്റകുറ്റ കുറ്റകുറ്റ. Technical terms English-കുറ്റ കുറ്റകുറ്റ
കുറ്റകുറ്റകുറ്റകുറ്റ.

Categorization and Clustering Method.

Categorization and Clustering Method. is an official topic in Module 4 of Data Mining. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Categorization and Clustering Method. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, categorization and clustering method. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Categorization and Clustering Method. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-☐☐ Categorization and Clustering Method.

日本語の定義/意味を説明してください。 日本語の定義/意味を説明してください, steps, application 日本語の定義/意味を説明してください. Technical terms English-日本語の定義/意味を説明してください.

Partitioning Method: K-Means 2 Apply Hierarchical Clustering method: Agglomerative

Partitioning Method: K-Means 2 Apply Hierarchical Clustering method: Agglomerative is an official topic in Module 4 of Data Mining. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Partitioning Method: K-Means 2 Apply Hierarchical Clustering method: Agglomerative using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, partitioning method: k-means 2 apply hierarchical clustering method: agglomerative should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Partitioning Method: K-Means 2 Apply Hierarchical Clustering method: Agglomerative means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-പാർട്ടിഷണിംഗ് മെഥോഡ്: K-Means 2 Apply
Hierarchical Clustering method: Agglomerative പാർട്ടിഷണിംഗ് മെഥോഡ്
definition/meaning പാർട്ടിഷണിംഗ് മെഥോഡ്. പാർട്ടിഷണിംഗ് മെഥോഡ്, steps, application
പാർട്ടിഷണിംഗ് മെഥോഡ് പാർട്ടിഷണിംഗ്. Technical terms English-പാർട്ടിഷണിംഗ് മെഥോഡ് പാർട്ടിഷണിംഗ്.

2 Understanding hierarchical clustering

2 Understanding hierarchical clustering is an official topic in Module 4 of Data Mining. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding hierarchical clustering using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding hierarchical clustering should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding hierarchical clustering means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-2 Understanding hierarchical clustering
ഈ ഉപകരണത്തിന്റെ നിർവ്വചന/അർത്ഥം എന്താണ്. ഈ ഉപകരണത്തിന്റെ
steps, application എന്നിവയെക്കുറിച്ചും എഴുതുക. Technical terms English-ൽ
എഴുതുക.

Hierarchical Clustering method: BIRCH. 2 Understanding Association Rule Mining: Concepts- support and confidence, FP-Growth Algorithm- Cluster Analysis-Introduction, Concepts, Types of Data in cluster analysis, Categorization and Clustering Method. Partitioning Method: K-Means an K-Medoid Clustering. Hierarchical Clustering method: Agglomerative hierarchical clustering, BIRCH. Text / Reference T/R Book Title/Author T1 Dunham M H, “Data Mining: Introductory and Advanced Topics”, Pearson Education, New Delhi, 2003. T2 Jaiwei Han and Micheline Kamber, “Data Mining Concepts and Techniques”, Elsevier, 2006. Online Resources Sl.No Website Link 1 <https://www.investopedia.com/terms/d/datamining.asp> 2 <https://www.techtarget.com/> 3 <https://www.javatpoint.com/data-mining> 4 <https://www.electronicshub.org/artificial-neural-networks-ann/> 5 https://www.tutorialspoint.com/data_mining/ 6 <https://towardsdatascience.com/the-fp-growth-algorithm>

Hierarchical Clustering method: BIRCH. 2 Understanding Association Rule Mining: Concepts- support and confidence, FP-Growth Algorithm- Cluster Analysis- Introduction, Concepts, Types of Data in cluster analysis, Categorization and Clustering Method. Partitioning Method: K-Means an K-Medoid Clustering. Hierarchical Clustering method: Agglomerative hierarchical clustering, BIRCH. Text / Reference T/R Book Title/Author T1 Dunham M H, “Data Mining: Introductory and Advanced Topics”, Pearson Education, New Delhi, 2003. T2 Jaiwei Han and Micheline Kamber, “Data Mining Concepts and Techniques”, Elsevier, 2006. Online Resources Sl.No Website Link 1 <https://www.investopedia.com/terms/d/datamining.asp> 2 <https://www.techtarget.com/> 3 <https://www.javatpoint.com/data-mining> 4 <https://www.electronicshub.org/artificial-neural-networks-ann/> 5 https://www.tutorialspoint.com/data_mining/ 6 <https://towardsdatascience.com/the-fp-growth-algorithm> is an official topic in Module 4 of Data Mining. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Hierarchical Clustering method: BIRCH. 2 Understanding Association Rule Mining: Concepts- support and confidence, FP-Growth Algorithm- Cluster Analysis-Introduction, Concepts, Types of Data in cluster analysis, Categorization and Clustering Method. Partitioning Method: K-Means and K-Medoid Clustering. Hierarchical Clustering method: Agglomerative hierarchical clustering, BIRCH. Text / Reference T/R Book Title/Author T1 Dunham M H, "Data Mining: Introductory and Advanced Topics", Pearson Education, New Delhi, 2003. T2 Jaiwei Han and Micheline Kamber, "Data Mining Concepts and Techniques", Elsevier, 2006. Online Resources Sl.No Website Link 1 <https://www.investopedia.com/terms/d/datamining.asp> 2 <https://www.techtarget.com/> 3 <https://www.javatpoint.com/data-mining> 4 <https://www.electronicshub.org/artificial-neural-networks-ann/> 5 https://www.tutorialspoint.com/data_mining/ 6 <https://towardsdatascience.com/the-fp-growth-algorithm-using-standard-technical-terminology>.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, hierarchical clustering method: birch. 2 understanding association rule mining: concepts- support and confidence, fp-growth algorithm- cluster analysis-introduction, concepts, types of data in cluster analysis, categorization and clustering method. partitioning method: k-means and k-medoid clustering. hierarchical clustering method: agglomerative hierarchical clustering, birch. text / reference t/r book title/author t1 dunham m h, "data mining: introductory and advanced topics", pearson education, new delhi, 2003. t2 jaiwei han and micheline kamber, "data mining concepts and techniques", elsevier, 2006. online resources sl.no website link 1 <https://www.investopedia.com/terms/d/datamining.asp> 2 <https://www.techtarget.com/> 3 <https://www.javatpoint.com/data-mining> 4 <https://www.electronicshub.org/artificial-neural-networks-ann/> 5 https://www.tutorialspoint.com/data_mining/ 6 <https://towardsdatascience.com/the-fp-growth-algorithm> should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Hierarchical Clustering method: BIRCH. 2 Understanding Association Rule Mining: Concepts- support and confidence, FP-Growth Algorithm- Cluster Analysis-Introduction, Concepts, Types of Data in cluster analysis, Categorization and Clustering Method. Partitioning Method: K-Means and K-Medoid Clustering. Hierarchical Clustering method: Agglomerative hierarchical clustering, BIRCH. Text / Reference T/R Book Title/Author T1 Dunham M H, "Data Mining: Introductory and Advanced Topics", Pearson Education, New Delhi, 2003. T2 Jaiwei Han and Micheline Kamber, "Data Mining Concepts and Techniques", Elsevier, 2006. Online Resources Sl.No Website Link 1

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

:				
Select appropriate clustering and algorithms for various applications,				
M4.01	Association Rule Mining: Concepts- support and confidence	1	Apply	
M4.02	FP-Growth Algorithm	3	Apply	
M4.03	Cluster Analysis-Introduction, Concepts Understanding	1		
M4.04	Categorization and Clustering Method. Understanding	2		
M4.05	Partitioning Method: K-Means Hierarchical Clustering method: Agglomerative	2	Apply	
M4.06	Understanding hierarchical clustering	2		
M4.07	Hierarchical Clustering method: BIRCH. Understanding	2		
	Series Test – II	1		
Contents:				
Association Rule Mining: Concepts- support and confidence, FP-Growth Algorithm-Cluster Analysis-Introduction, Concepts, Types of Data in cluster analysis, Categorization and Clustering Method.				
Partitioning Method: K-Means an K-Medoid Clustering.				
Hierarchical Clustering method: Agglomerative hierarchical clustering, BIRCH.				

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

Definition → Principle → Components/steps → Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure → Observation → Result → Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain Data Mining Concepts and Applications. (3 marks)

Answer: Begin with the standard definition or identity of *Data Mining Concepts and Applications*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Data Mining Stages. (3 marks)

Answer: Begin with the standard definition or identity of *Data Mining Stages*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Data Mining Models. (3 marks)

Answer: Begin with the standard definition or identity of *Data Mining Models*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Applications of Data Mining. (3 marks)

Answer: Begin with the standard definition or identity of *Applications of Data Mining*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Data Preprocessing Concepts 1 Understanding.. (3 marks)

Answer: Begin with the standard definition or identity of *Data Preprocessing Concepts 1 Understanding..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Data Cleaning. (3 marks)

Answer: Begin with the standard definition or identity of *Data Cleaning*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Data Integration 1 Applying Data Reduction Strategies Overview-Dimensionality. (3 marks)

Answer: Begin with the standard definition or identity of *Data Integration 1 Applying Data Reduction Strategies Overview-Dimensionality*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 2 Understanding. Reduction, Numerosity Reduction Data Transformation Strategies Overview-smoothing,. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding. Reduction, Numerosity Reduction Data Transformation Strategies Overview-smoothing,.* State the

governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 3

Define or explain 2 Understanding characteristics. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding characteristics*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Back propagation Algorithm.. (3 marks)

Answer: Begin with the standard definition or identity of *Back propagation Algorithm..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Support Vector Machines 2 Understanding.. (3 marks)

Answer: Begin with the standard definition or identity of *Support Vector Machines 2 Understanding..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain Lazy Learners-K Nearest Neighbor Classifier.. (3 marks)

Answer: Begin with the standard definition or identity of *Lazy Learners-K Nearest Neighbor Classifier..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Module 4

Define or explain 1 Apply confidence. (3 marks)

Answer: Begin with the standard definition or identity of *1 Apply confidence.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain FP-Growth Algorithm 3 Apply. (3 marks)

Answer: Begin with the standard definition or identity of *FP-Growth Algorithm 3 Apply.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain Cluster Analysis-Introduction, Concepts. (3 marks)

Answer: Begin with the standard definition or identity of *Cluster Analysis-Introduction, Concepts*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Categorization and Clustering Method.. (3 marks)

Answer: Begin with the standard definition or identity of *Categorization and Clustering Method..*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Data Mining Concepts and Applications**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Data Preprocessing Concepts 1 Understanding..**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **2 Understanding characteristics.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **1 Apply confidence.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link
- <https://www.investopedia.com/terms/d/datamining.asp> <https://www.techtarget.com/>
<https://www.javatpoint.com/data-mining>

- <https://www.electronicshub.org/artificial-neural-networks-ann/>
https://www.tutorialspoint.com/data_mining/
- <https://towardsdatascience.com/the-fp-growth-algorithm>

IN-PAGE SEARCH

Search this lesson

Prepared from the supplied official SITTTTR syllabus PDF for Course 5302. Verify institutional updates against the current official curriculum.